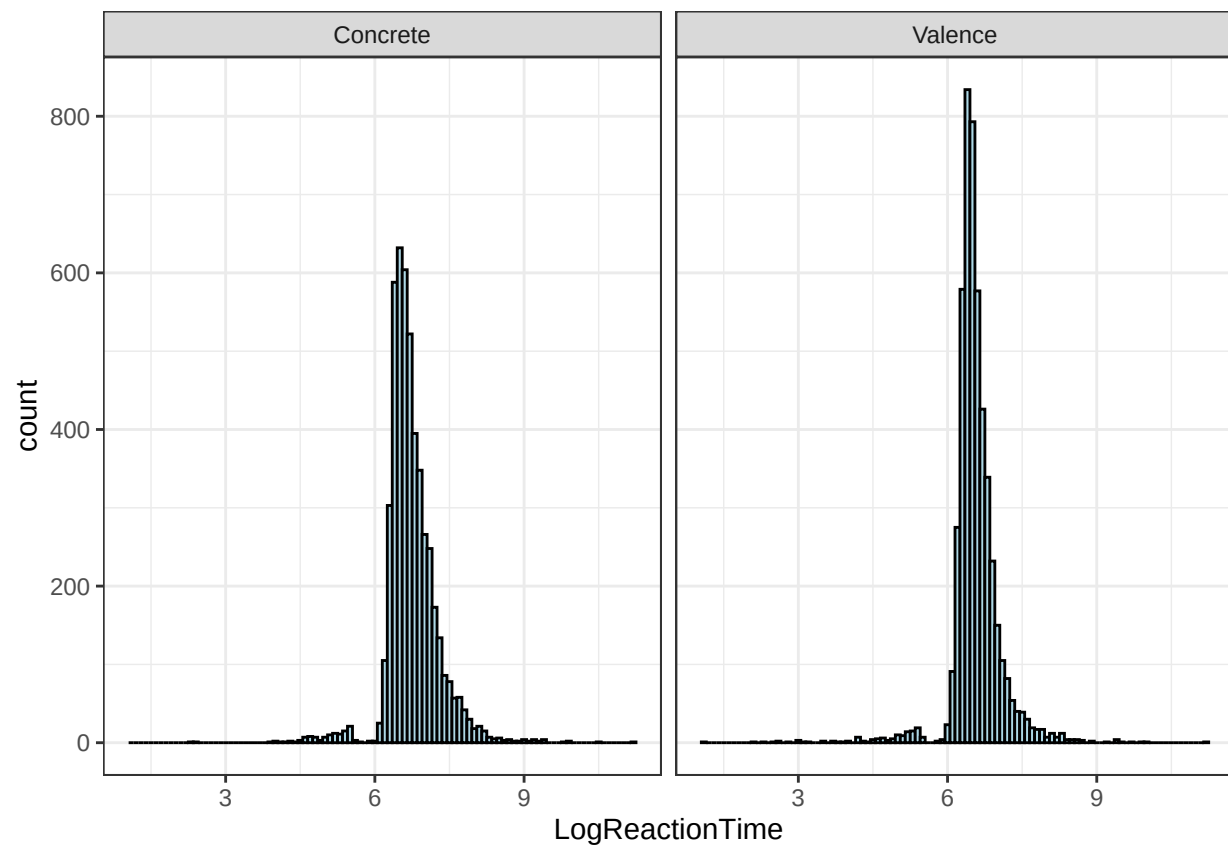


Nouns Conc-Abs: Analysis ReactionTime

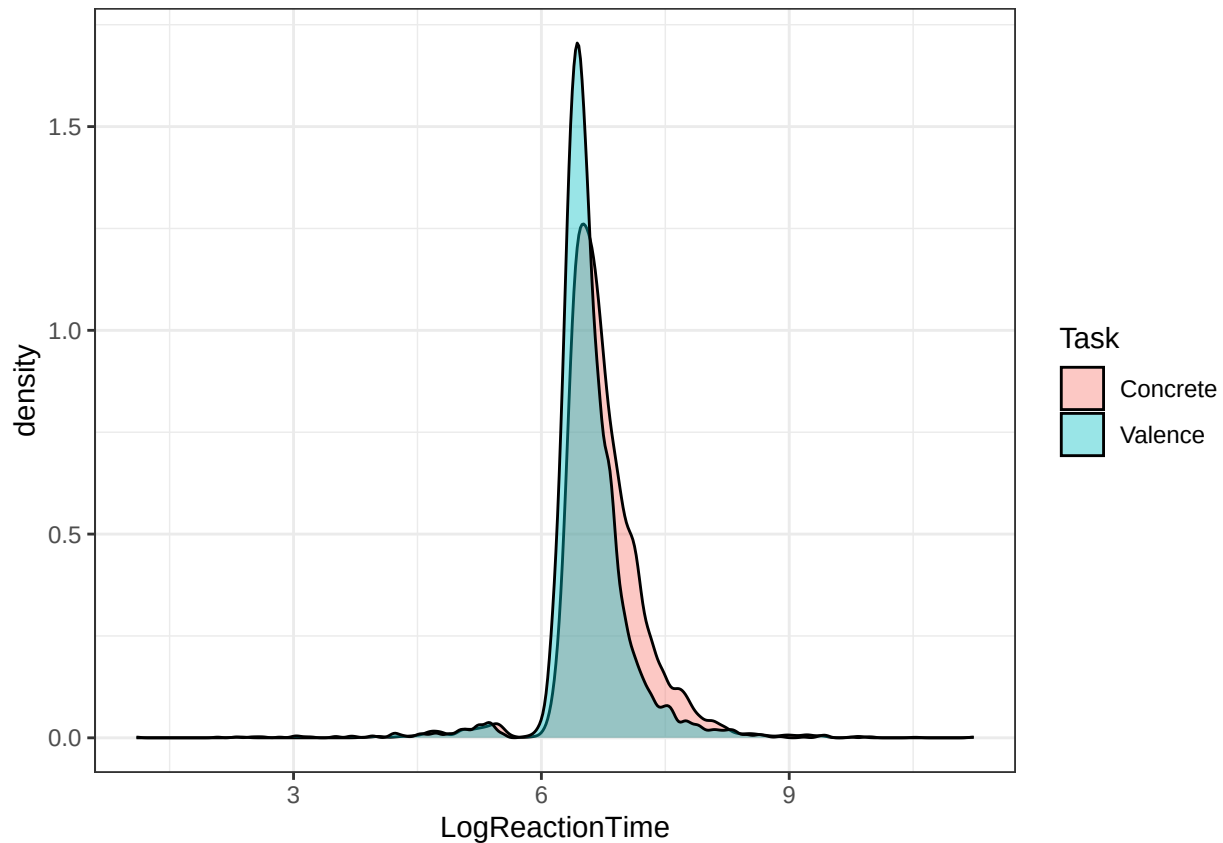
author

2026-05-29

```
ggplot(d, aes(x=LogReactionTime)) +  
  geom_histogram(binwidth = .1, fill = "lightblue", color = "black") +  
  facet_wrap(~Task)
```



```
ggplot(d, aes(x=LogReactionTime, fill=Task)) +  
  geom_density(alpha = .4)
```



```
names(d)
```

```
## [1] "X"           "ID.true"      "Word"         "Label"
## [5] "ConcValCombo" "Task"         "BlockOrder"   "Group"
## [9] "Response"     "Accuracy"     "EventTime"    "Value"
## [13] "RT"           "ReactionTime" "Key_value_F"  "Key_value_J"
## [17] "Comments"     "LogReactionTime" "LogRT"        "TrialNumber"
```

```
dcen <- d %>%
```

```
  mutate(Word = as.factor(Word),
         ID.true = as.factor(ID.true),
         Task = as.factor(Task),
         cAccuracy = as.numeric(Accuracy) - mean(as.numeric(Accuracy)),
         cTask = as.numeric(Task) - mean(as.numeric(Task)))
```

```
m <- lmer(LogReactionTime ~ cAccuracy*cTask + (1+cTask|Word) + (1+cTask|ID.true), data = dcen)
summary(m)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: LogReactionTime ~ cAccuracy * cTask + (1 + cTask | Word) + (1 +
##       cTask | ID.true)
## Data: dcen
##
## REML criterion at convergence: 9610.1
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
```

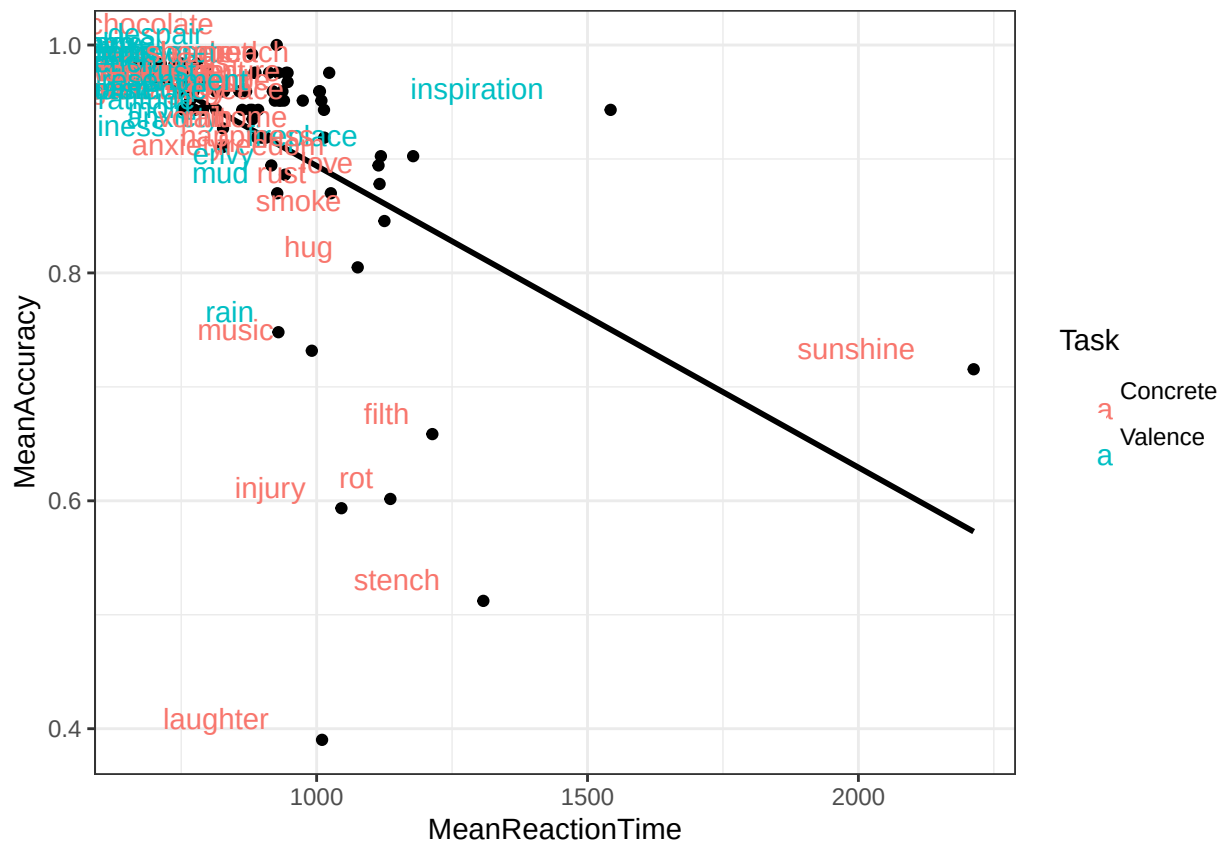
```
## -11.9217 -0.5415 -0.1484 0.3698 11.4676
##
## Random effects:
## Groups Name Variance Std.Dev. Corr
## Word (Intercept) 0.001831 0.04279
## cTask 0.016931 0.13012 -0.18
## ID.true (Intercept) 0.110705 0.33272
## cTask 0.020169 0.14202 0.00
## Residual 0.148540 0.38541
## Number of obs: 9840, groups: Word, 40; ID.true, 40
##
## Fixed effects:
## Estimate Std. Error df t value Pr(>|t|)
## (Intercept) 6.650e+00 5.319e-02 4.025e+01 125.033 < 2e-16 ***
## cAccuracy -2.909e-03 1.693e-02 9.603e+03 -0.172 0.864
## cTask -1.777e-01 3.147e-02 6.854e+01 -5.646 3.42e-07 ***
## cAccuracy:cTask 3.184e-02 3.364e-02 9.214e+03 0.947 0.344
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
## (Intr) cAccrc cTask
## cAccuracy -0.004
## cTask -0.015 -0.039
## cAccrcy:cTs -0.012 0.347 -0.014

agr <- d %>%
  group_by(Word,Task) %>%
  summarize(MeanAccuracy = mean(Accuracy),
            MeanReactionTime = mean(ReactionTime))

## `summarise()` has grouped output by 'Word'. You can override using the
## `.groups` argument.

ggplot(agr, aes(x = MeanReactionTime, y = MeanAccuracy)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE, color = "black") +
  geom_text(aes(label = Word, color = Task), vjust = -0.5, hjust = 1.5)

## `geom_smooth()` using formula = 'y ~ x'
```



```
# guides(legend = "none")
# theme(legend.position = "none") # Remove the legend
# ggsave("../graphs/nouns2_accXrt.pdf",width = 5, height = 3)

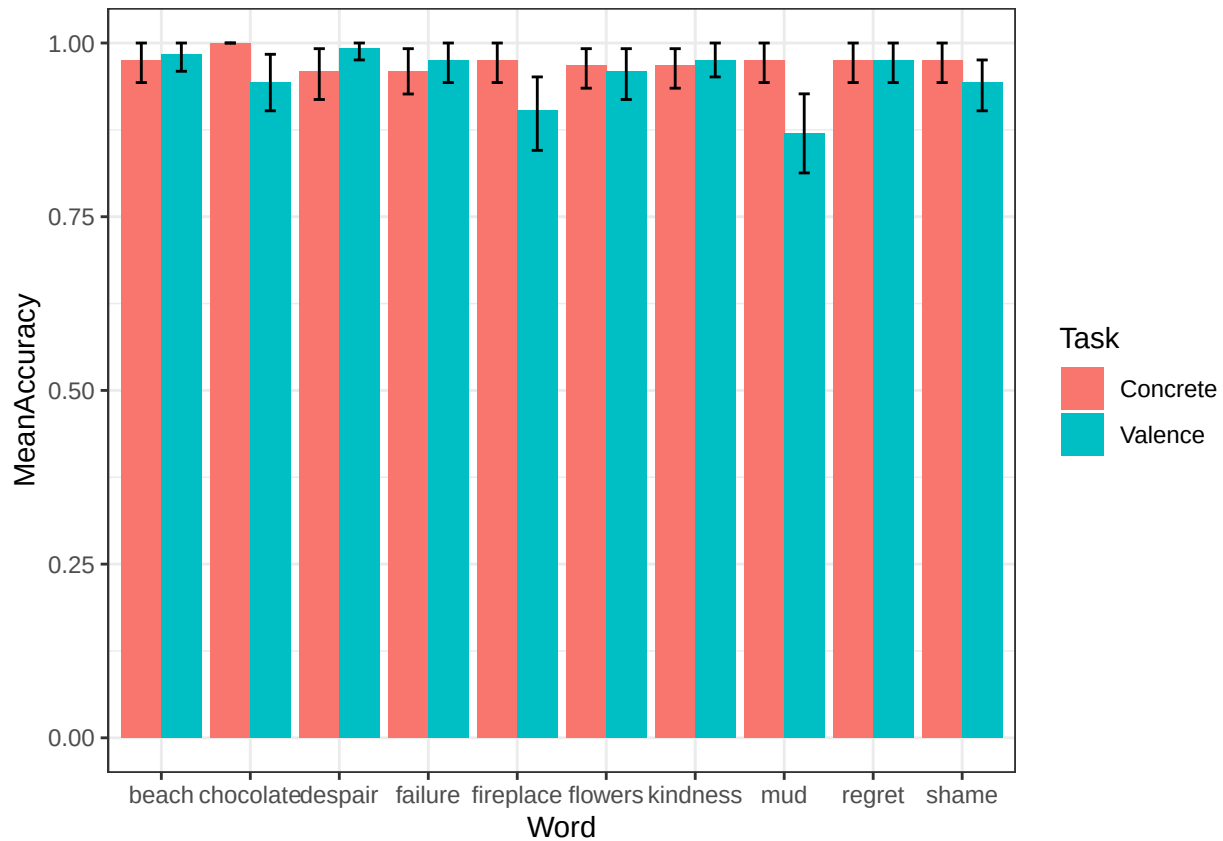
# Compute highest accuracy for Concrete
concrete_accuracy <- d %>%
  group_by(Word,Task) %>%
  summarize(MeanAccuracy = mean(Accuracy),
            MeanReactionTime = mean(ReactionTime)) %>%
  filter(Task == "Concrete") %>%
  select(Word, MeanAccuracy) %>%
  rename(ConcreteAccuracy = MeanAccuracy) %>%
  arrange(desc(ConcreteAccuracy)) %>%
  head(10)

## `summarise()` has grouped output by 'Word'. You can override using the
## `.groups` argument.

# View(concrete_accuracy)

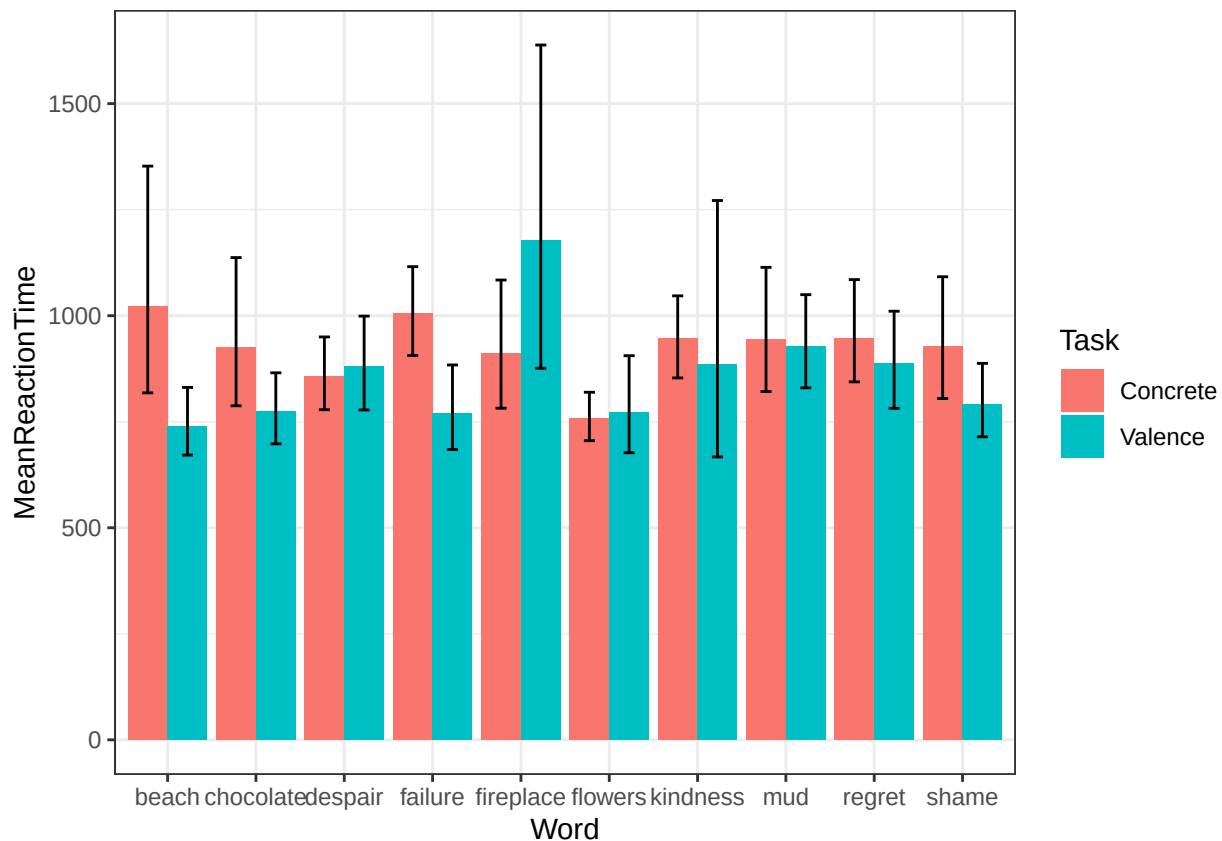
agr <- d %>%
  filter(Word %in% concrete_accuracy$Word) %>%
  group_by(Word,Task) %>%
  summarize(MeanAccuracy = mean(Accuracy),
            MeanReactionTime = mean(ReactionTime))

## `summarise()` has grouped output by 'Word'. You can override using the
## `.groups` argument.
```

```
agr <- d %>%
  filter(Word %in% concrete_accuracy$Word) %>%
  group_by(Word, Task) %>%
  reframe(MeanReactionTime = mean(ReactionTime),
           CILow = ci.low(ReactionTime),
           CIHigh = ci.high(ReactionTime)) %>%
  mutate(YMin = MeanReactionTime - CILow,
         YMax = MeanReactionTime + CIHigh)
# View(agr)

dodge = position_dodge(.9)
ggplot(data=agr, aes(x=Word, y=MeanReactionTime, fill = Task)) +
  geom_bar(position=dodge, stat="identity") +
  geom_errorbar(aes(ymin=YMin, ymax=YMax), width=.25, position=position_dodge(0.9))
```



First Remove participants who aren't super , aggregating over Task

```
length(unique(d$ID.true))
```

```
## [1] 40
```

```
inacc.parts <- d %>%
  group_by(ID.true, Task) %>%
  summarise(MeanAccuracy = mean(Accuracy)) %>%
  filter(MeanAccuracy < .75)
```

```
## `summarise()` has grouped output by 'ID.true'. You can override using the
## `.groups` argument.
```

```
# How many participants have Accuracy < .75?
length(unique(inacc.parts$ID.true))
```

```
## [1] 2
```

```
d.inaccurate.removed <- d %>%
  anti_join(inacc.parts, by = "ID.true")

# Sanity check
length(unique(d.inaccurate.removed$ID.true))
```

```
## [1] 38
```

remove all inaccurate trials

```
orig <- nrow(d.inaccurate.removed)
d.inaccurate.removed <- d.inaccurate.removed %>%
  filter(Accuracy == 1)
nrow(d.inaccurate.removed)/orig*100
```

```
## [1] 92.87393
```

```
# Remove subjects with ReactionTime higher than 3x IQR
summary(d.inaccurate.removed$LogReactionTime)
```

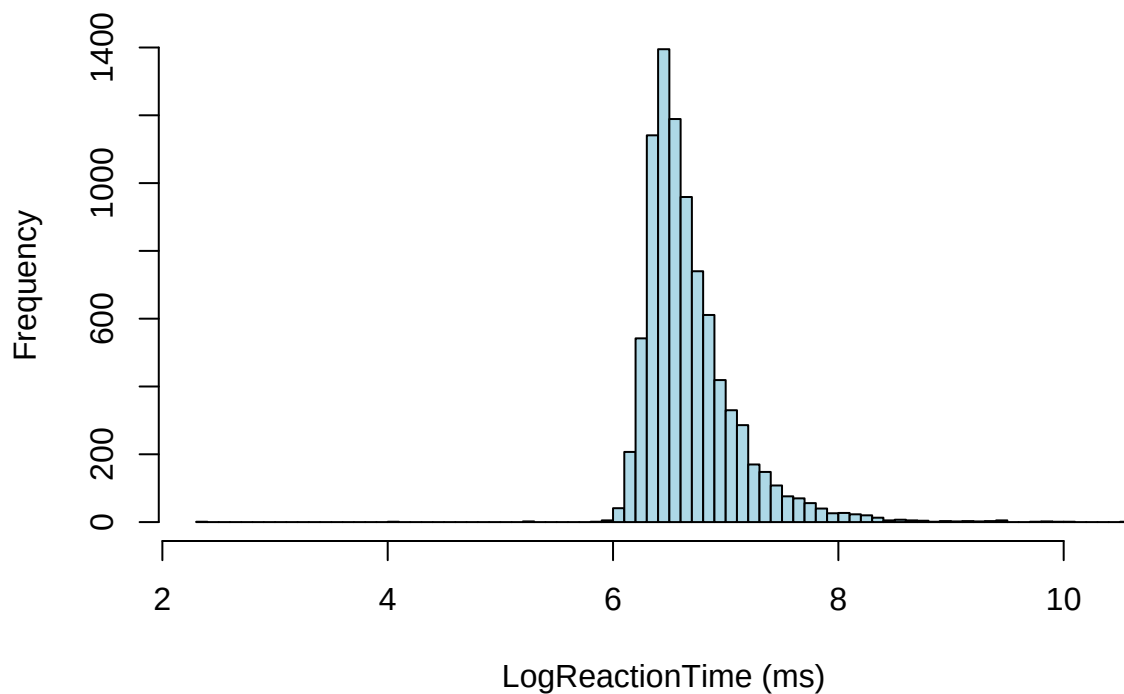
```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      2.303   6.417   6.583   6.685   6.844   10.528
```

```
# Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
# 6.924   7.328   7.436   7.479   7.579   10.008
range(d.inaccurate.removed$LogReactionTime)
```

```
## [1] 2.302585 10.527553
```

```
hist(d.inaccurate.removed$LogReactionTime, breaks=100, col="lightblue", xlab="LogReactionTime (ms)",
     main="Histogram with Normal Curve")
```

Histogram with Normal Curve



```
quantile(d.inaccurate.removed$LogReactionTime)
```

```
##           0%           25%           50%           75%          100%
## 2.302585  6.416732  6.583409  6.843750 10.527553
```

```
IQR(d.inaccurate.removed$LogReactionTime)*3 # 0.7526289
```

```
## [1] 1.281053
```



```

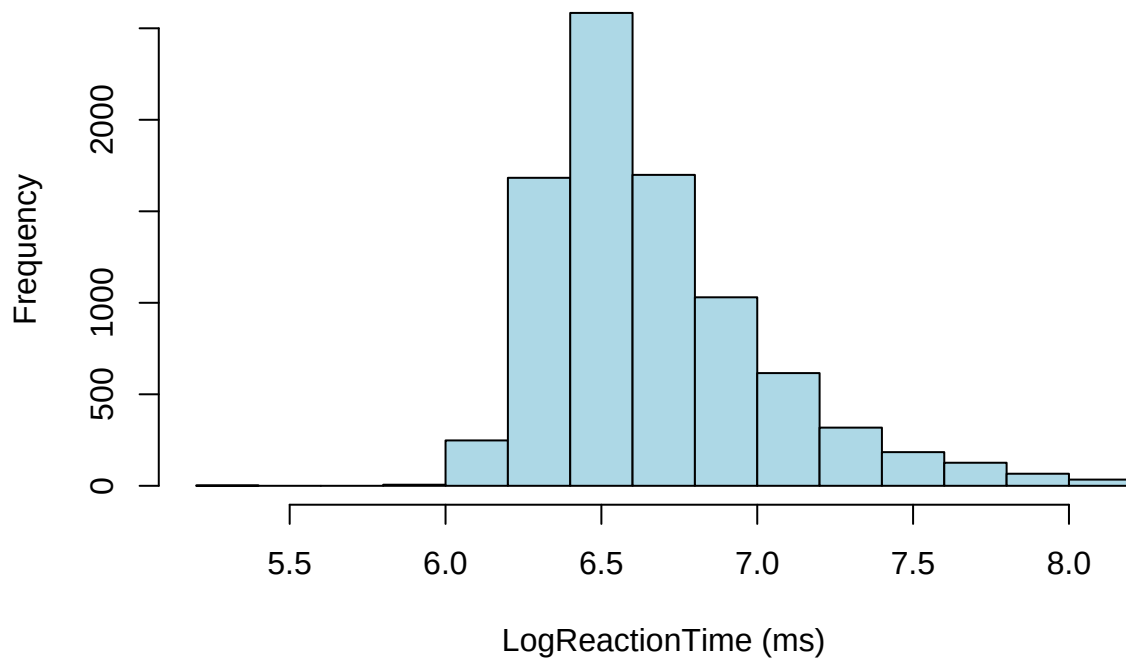
cutoff.high <- quantile(d.inaccurate.removed$LogReactionTime)[4] + IQR(d.inaccurate.removed$LogReactionTime)
cutoff.low <- quantile(d.inaccurate.removed$LogReactionTime)[2] - IQR(d.inaccurate.removed$LogReactionTime)

# remove subjects with ReactionTime higher than 3 x IQR
df.outliers.removed <- subset(d.inaccurate.removed, (d.inaccurate.removed$LogReactionTime > cutoff.low)

hist(df.outliers.removed$LogReactionTime, col="lightblue", xlab="LogReactionTime (ms)",
     main="Histogram with Normal Curve")

```

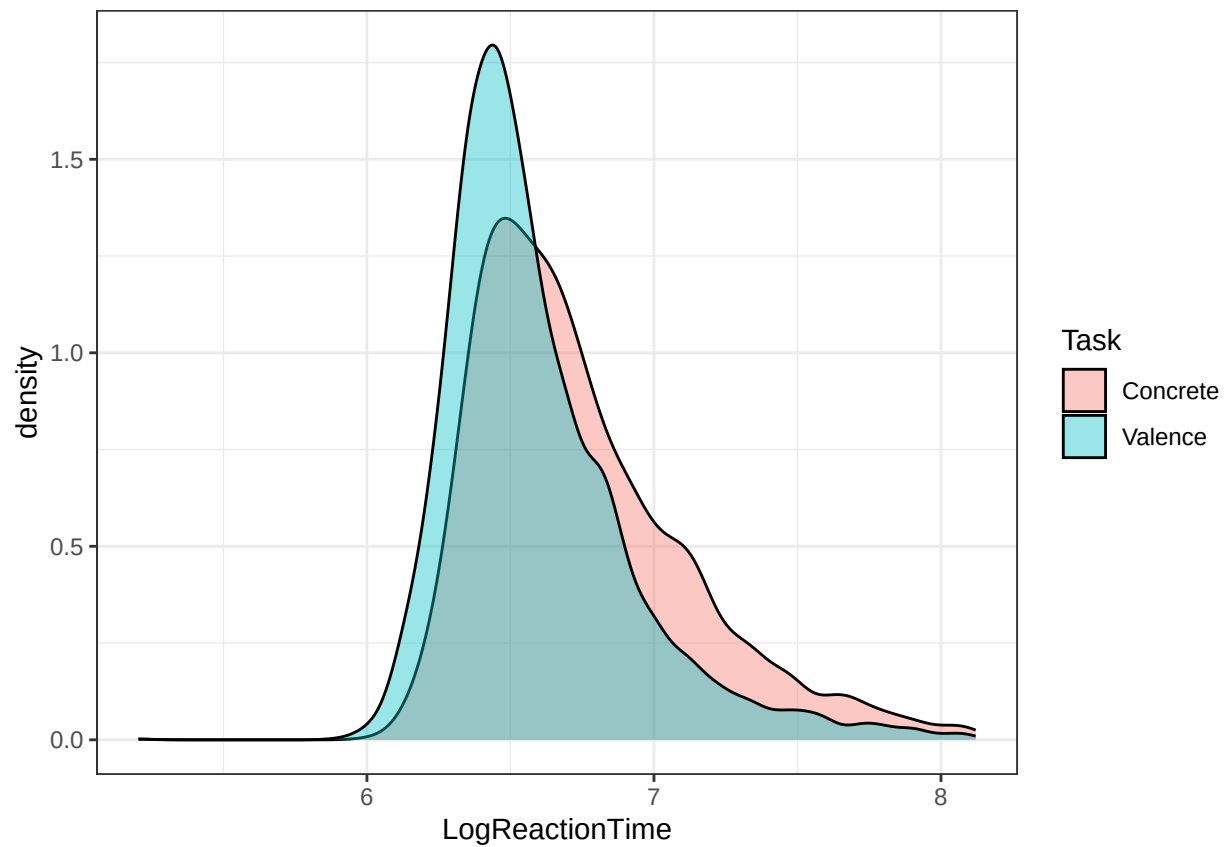
Histogram with Normal Curve



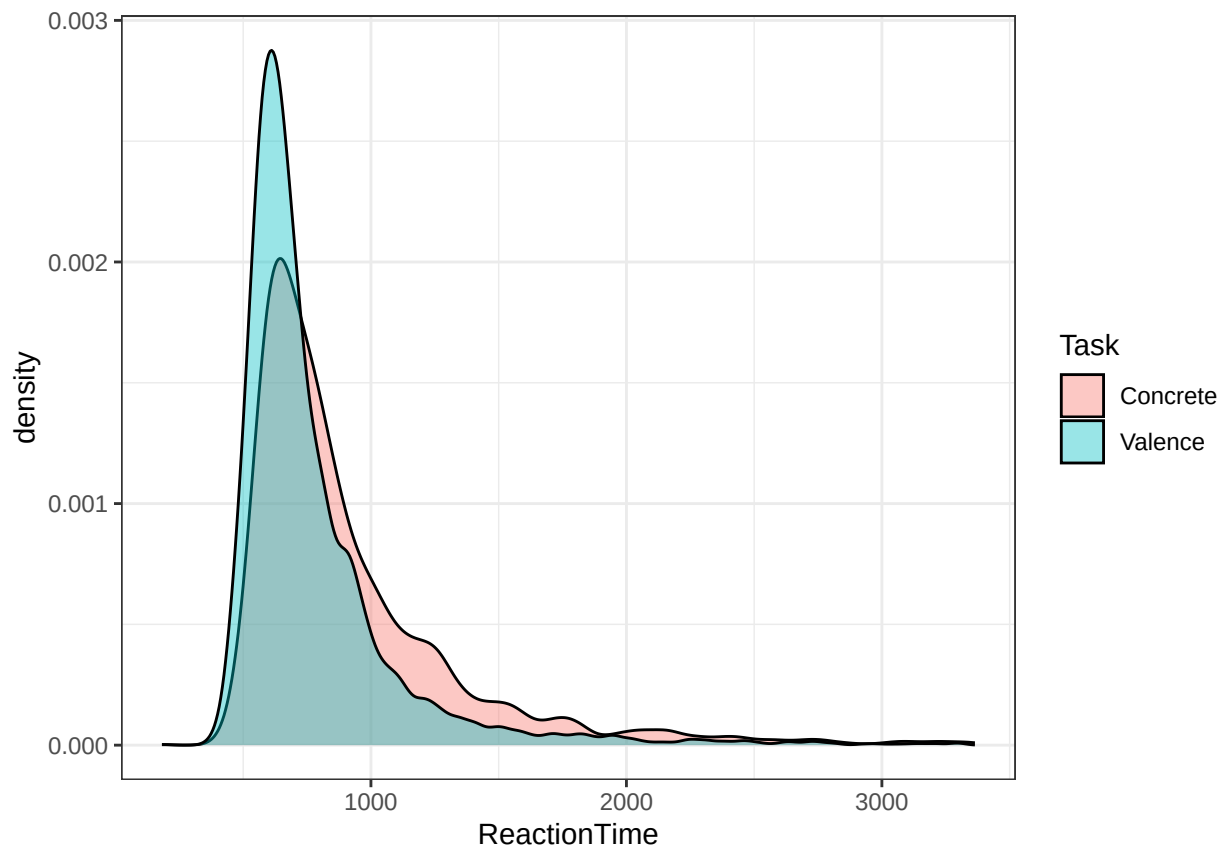
```

ggplot(df.outliers.removed, aes(x=LogReactionTime, fill=Task)) +
  # facet_wrap(~BlockOrder) +
  geom_density(alpha = .4)

```



```
ggplot(df.outliers.removed, aes(x=ReactionTime, fill=Task)) +  
  # facet_wrap(~BlockOrder) +  
  geom_density(alpha = .4)
```



convert everything to factors

Is there a difference between Semantic and Valence Tasks?

Yes

```
m = lmer(LogReactionTime ~ cTask + (1+cTask|ID.true) + (1+cTask|Word), data=center)
summary(m)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: LogReactionTime ~ cTask + (1 + cTask | ID.true) + (1 + cTask |
##      Word)
##      Data: center
##
## REML criterion at convergence: 3815
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -5.6966 -0.6296 -0.1712  0.4579  4.9763
##
## Random effects:
##      Groups    Name                Variance Std.Dev. Corr
##      Word      (Intercept)  0.002426  0.04925
##              cTask        0.015390  0.12406  -0.35
##      ID.true   (Intercept)  0.025989  0.16121
```

```
##           cTask           0.018896 0.13746 -0.34
## Residual           0.086842 0.29469
## Number of obs: 8596, groups: Word, 40; ID.true, 38
##
## Fixed effects:
##           Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  6.67343    0.02748 43.28457 242.868 < 2e-16 ***
## cTask       -0.16943    0.03040 67.39518 -5.574 4.74e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr)
## cTask -0.304
```

Is there an Interaction between Task and WordType (ConcVal-Combo)?

Y.

```
m = lmer(LogReactionTime ~ cTask*ConcValCombo + (1+ConcValCombo+cTask|ID.true) + (1+cTask|Word), data=c
```

```
## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, :
## Model failed to converge with max|grad| = 0.00319275 (tol = 0.002, component 1)
```

```
saveRDS(m, "../models/model-Task-ConcValCombo_outlier_excl_ReactionTime.rds")
```

```
# m <- readRDS("../models/model-Task-ConcValCombo_outlier_excl_ReactionTime.rds")
summary(m)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: LogReactionTime ~ cTask * ConcValCombo + (1 + ConcValCombo +
##           cTask | ID.true) + (1 + cTask | Word)
## Data: center
##
## REML criterion at convergence: 3782.2
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -5.4796 -0.6263 -0.1720  0.4528  4.9739
##
## Random effects:
## Groups   Name                                Variance Std.Dev. Corr
## Word     (Intercept)                        0.001906 0.04366
##           cTask                             0.013804 0.11749 -0.37
## ID.true  (Intercept)                        0.028097 0.16762
##           ConcValComboabstract-positive    0.002085 0.04567 -0.44
##           ConcValComboconcrete-negative    0.002275 0.04769  0.11  0.14
##           ConcValComboconcrete-positive    0.004562 0.06754 -0.32  0.51  0.77
##           cTask                             0.018893 0.13745 -0.32 -0.23  0.03
## Residual                                0.085746 0.29282
##
##
```

```

##
##
##
##
## 0.04
##
## Number of obs: 8596, groups: Word, 40; ID.true, 38
##
## Fixed effects:
##
## Estimate Std. Error df t value
## (Intercept) 6.67242 0.03113 52.25312 214.363
## cTask -0.08248 0.04508 54.08582 -1.830
## ConcValComboabstract-positive -0.01989 0.02266 39.64024 -0.878
## ConcValComboconcrete-negative 0.04753 0.02288 40.76562 2.077
## ConcValComboconcrete-positive -0.02013 0.02415 46.35046 -0.834
## cTask:ConcValComboabstract-positive -0.12481 0.05540 34.36253 -2.253
## cTask:ConcValComboconcrete-negative -0.14585 0.05559 34.83238 -2.624
## cTask:ConcValComboconcrete-positive -0.08041 0.05556 34.74630 -1.447
## Pr(>|t|)
## (Intercept) <2e-16 ***
## cTask 0.0728 .
## ConcValComboabstract-positive 0.3854
## ConcValComboconcrete-negative 0.0441 *
## ConcValComboconcrete-positive 0.4088
## cTask:ConcValComboabstract-positive 0.0308 *
## cTask:ConcValComboconcrete-negative 0.0128 *
## cTask:ConcValComboconcrete-positive 0.1568
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
## (Intr) cTask CncVlCmbb- CncVlCmbcnrt-n CncVlCmbcnrt-p
## cTask -0.270
## CncVlCmbb- -0.450 0.146
## CncVlCmbcnrt-n -0.291 0.187 0.457
## CncVlCmbcnrt-p -0.432 0.181 0.496 0.534
## cTsk:CncVlCmbb- 0.108 -0.614 -0.300 -0.148 -0.140
## cTsk:CncVlCmbcnrt-n 0.108 -0.612 -0.149 -0.303 -0.140
## cTsk:CncVlCmbcnrt-p 0.108 -0.613 -0.149 -0.147 -0.285
## cTsk:CncVlCmbb- cTsk:CncVlCmbcnrt-n
## cTask
## CncVlCmbb-
## CncVlCmbcnrt-n
## CncVlCmbcnrt-p
## cTsk:CncVlCmbb-
## cTsk:CncVlCmbcnrt-n 0.498
## cTsk:CncVlCmbcnrt-p 0.499 0.497
## optimizer (nloptwrap) convergence code: 0 (OK)
## Model failed to converge with max|grad| = 0.00319275 (tol = 0.002, component 1)

```

Does Accuracy predict reaction time?

In other words, is reaction time affected by certainty about the categorization? - No.

```
m = lmer(LogReactionTime ~ cAccuracy + (1|ID.true) + (1|Word), data=center)

## fixed-effect model matrix is rank deficient so dropping 1 column / coefficient

summary(m)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: LogReactionTime ~ cAccuracy + (1 | ID.true) + (1 | Word)
## Data: center
##
## REML criterion at convergence: 4917.7
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -5.0713 -0.6478 -0.1866  0.4659  4.6172
##
## Random effects:
## Groups Name Variance Std.Dev.
## Word (Intercept) 0.001606 0.04007
## ID.true (Intercept) 0.025535 0.15980
## Residual 0.101170 0.31807
## Number of obs: 8596, groups: Word, 40; ID.true, 38
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  6.66728    0.02691 41.20476   247.8   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## fit warnings:
## fixed-effect model matrix is rank deficient so dropping 1 column / coefficient
```

Main Effect of Block Order

On ReactionTime

- No.

```
m = lmer(LogReactionTime ~ cBlockOrder + (1|ID.true) + (1+cBlockOrder|Word), data=center)

summary(m)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: LogReactionTime ~ cBlockOrder + (1 | ID.true) + (1 + cBlockOrder |
## Word)
## Data: center
##
## REML criterion at convergence: 4917
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
```

```
## -5.0798 -0.6481 -0.1844  0.4652  4.6384
##
## Random effects:
##   Groups   Name                Variance Std.Dev. Corr
##   Word      (Intercept) 1.603e-03 0.040039
##             cBlockOrder 9.592e-05 0.009794 -0.67
##   ID.true   (Intercept) 2.346e-02 0.153157
##   Residual                1.011e-01 0.318037
## Number of obs: 8596, groups: Word, 40; ID.true, 38
##
## Fixed effects:
##               Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   6.66561    0.02588 40.43042 257.515  <2e-16 ***
## cBlockOrder  -0.10349    0.05026 36.02080  -2.059   0.0468 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr)
## cBlockOrder 0.026
```

effect of ConcValCombo on ReactionTime?

nope.

```
m = lmer(LogReactionTime ~ ConcValCombo + (1+ConcValCombo|ID.true) + (1|Word), data=center)
```

```
## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, :
## Model failed to converge with max|grad| = 0.0198304 (tol = 0.002, component 1)
```

```
summary(m)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: LogReactionTime ~ ConcValCombo + (1 + ConcValCombo | ID.true) +
##          (1 | Word)
## Data: center
##
## REML criterion at convergence: 4901.1
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.9521 -0.6444 -0.1862  0.4637  4.6411
##
## Random effects:
##   Groups   Name                Variance Std.Dev. Corr
##   Word      (Intercept)      0.001129 0.03360
##   ID.true   (Intercept)      0.027867 0.16693
##             ConcValComboabstract-positive 0.001476 0.03841 -0.51
##             ConcValComboconcrete-negative 0.001567 0.03958  0.13 -0.10
##             ConcValComboconcrete-positive 0.003105 0.05573 -0.37  0.63  0.69
##   Residual                0.100394 0.31685
## Number of obs: 8596, groups: Word, 40; ID.true, 38
##
## Fixed effects:
```

```
##               Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      6.67346    0.02986 46.32498 223.500  <2e-16 ***
## ConcValComboabstract-positive -0.01991    0.01884 38.36429  -1.057    0.2972
## ConcValComboconcrete-negative  0.03154    0.01902 39.89565   1.658    0.1052
## ConcValComboconcrete-positive -0.03492    0.02005 42.92816  -1.742    0.0887 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) CncVlCmbb- CncVlCmbcncrt-n
## CncVlCmbbs-      -0.435
## CncVlCmbcncrt-n -0.238  0.429
## CncVlCmbcncrt-p -0.417  0.514      0.521
## optimizer (nloptwrap) convergence code: 0 (OK)
## Model failed to converge with max|grad| = 0.0198304 (tol = 0.002, component 1)
```

In the Concreteness task, is there a difference between concreteness and abstractness on ReactionTime?

- Nope

```
str(df_factors)
```

```
## 'data.frame': 8596 obs. of 20 variables:
## $ X : Factor w/ 8596 levels "2","3","4","5",...: 1 2 3 4 5 6 7 8 9 10 ...
## $ ID.true : Factor w/ 38 levels "5731e4005b8b06000f7e36b3",...: 2 2 2 2 2 2 2 2 2 2 ...
## $ Word : Factor w/ 40 levels "anxiety","beach",...: 3 39 14 12 7 13 19 25 9 6 ...
## $ Label : Factor w/ 2 levels "test_conc","test_val": 2 2 2 2 2 2 2 2 2 2 ...
## $ ConcValCombo : Factor w/ 4 levels "abstract-negative",...: 4 4 1 3 1 2 4 1 4 1 ...
## $ Task : Factor w/ 2 levels "Concrete","Valence": 2 2 2 2 2 2 2 2 2 2 ...
## $ BlockOrder : Factor w/ 2 levels "CV","VC": 2 2 2 2 2 2 2 2 2 2 ...
## $ Group : Factor w/ 2 levels "A","B": 1 1 1 1 1 1 1 1 1 1 ...
## $ Response : Factor w/ 4 levels "abstract","concrete",...: 4 4 3 3 3 4 4 3 4 3 ...
## $ Accuracy : Factor w/ 1 level "1": 1 1 1 1 1 1 1 1 1 1 ...
## $ EventTime : Factor w/ 8595 levels "1737993435128",...: 460 461 462 463 464 465 466 467 468 469 ...
## $ Value : Factor w/ 4 levels "abstract","concrete",...: 4 4 3 3 3 4 4 3 4 3 ...
## $ RT : Factor w/ 1324 levels "1267.666666666667",...: 165 386 71 107 22 306 50 442 160 11 ...
## $ ReactionTime : int 839 551 512 499 550 630 604 749 774 622 ...
## $ Key_value_F : Factor w/ 4 levels "abstract","concrete",...: 3 3 3 3 3 3 3 3 3 3 ...
## $ Key_value_J : Factor w/ 2 levels "A","B": 1 1 1 1 1 1 1 1 1 1 ...
## $ Comments : Factor w/ 0 levels: NA NA NA NA NA NA NA NA NA ...
## $ LogReactionTime: num 6.73 6.31 6.24 6.21 6.31 ...
## $ LogRT : Factor w/ 1324 levels "7.14493321926015",...: 165 386 71 107 22 306 50 442 160 11 ...
## $ TrialNumber : Factor w/ 448 levels "1","2","3","4",...: 2 3 4 5 6 7 8 9 10 11 ...
```

```
sem <- df_factors %>%
  filter(Task == "Concrete") %>%
  mutate(
    Semantic = ifelse(grepl("concrete", ConcValCombo), "concrete",
                     ifelse(grepl("abstract", ConcValCombo), "abstract", NA)),
    Valence = ifelse(grepl("positive", ConcValCombo), "positive",
                     ifelse(grepl("negative", ConcValCombo), "negative", NA)),
    cConcValCombo = as.numeric(ConcValCombo) - mean(as.numeric(ConcValCombo)),
    cSemantic = as.numeric(factor(Semantic)) - mean(as.numeric(factor(Semantic)))
```



```
)

m = lmer(LogReactionTime ~ cConcValCombo + (1+cConcValCombo|ID.true) + (1+cConcValCombo|Word), data=sem)
summary(m)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: LogReactionTime ~ cConcValCombo + (1 + cConcValCombo | ID.true) +
##          (1 + cConcValCombo | Word)
## Data: sem
##
## REML criterion at convergence: 2248.2
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -5.0697 -0.6449 -0.1719  0.4843  4.5761
##
## Random effects:
## Groups Name Variance Std.Dev. Corr
## Word (Intercept) 0.004098 0.06402
## cConcValCombo 0.004844 0.06960 0.95
## ID.true (Intercept) 0.038846 0.19709
## cConcValCombo 0.001217 0.03488 0.05
## Residual 0.094494 0.30740
## Number of obs: 4133, groups: Word, 40; ID.true, 38
##
## Fixed effects:
## Estimate Std. Error df t value Pr(>|t|)
## (Intercept) 6.77374 0.03479 47.35157 194.708 <2e-16 ***
## cConcValCombo 0.03704 0.01479 37.41334 2.505 0.0167 *
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
## (Intr)
## cConcValCmb 0.256
```

```
m = lmer(LogReactionTime ~ cSemantic + (1+cSemantic|ID.true) + (1|Word), data=sem)
summary(m)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: LogReactionTime ~ cSemantic + (1 + cSemantic | ID.true) + (1 |
## Word)
## Data: sem
##
## REML criterion at convergence: 2241.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -5.0929 -0.6439 -0.1747  0.4794  4.5607
##
## Random effects:
## Groups Name Variance Std.Dev. Corr
```

```
## Word      (Intercept) 0.009159 0.09570
## ID.true   (Intercept) 0.038833 0.19706
##          cSemantic    0.008903 0.09436 0.02
## Residual              0.093824 0.30631
## Number of obs: 4133, groups: Word, 40; ID.true, 38
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  6.76213    0.03572 51.88604 189.290 <2e-16 ***
## cSemantic     0.05201    0.03531 48.43900   1.473   0.147
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr)
## cSemantic -0.018
```

In the Valence task , is there a difference between positive and negative on ReactionTime?

- Nope.

```
val <- df_factors %>%
  filter(Task == "Valence") %>%
  mutate(
    Semantic = ifelse(grepl("concrete", ConcValCombo), "concrete",
                      ifelse(grepl("abstract", ConcValCombo), "abstract", NA)),
    Valence = ifelse(grepl("positive", ConcValCombo), "positive",
                     ifelse(grepl("negative", ConcValCombo), "negative", NA)),
    cConcValCombo = as.numeric(ConcValCombo) - mean(as.numeric(ConcValCombo)),
    cValence = as.numeric(factor(Valence)) - mean(as.numeric(factor(Valence)))
  )

m = lmer(LogReactionTime ~ cConcValCombo + (1+cConcValCombo|ID.true) + (1+cConcValCombo|Word), data=val)

## boundary (singular) fit: see help('isSingular')

summary(m)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: LogReactionTime ~ cConcValCombo + (1 + cConcValCombo | ID.true) +
##          (1 + cConcValCombo | Word)
## Data: val
##
## REML criterion at convergence: 1519.3
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -6.0649 -0.6137 -0.1668  0.4269  5.2469
##
## Random effects:
## Groups   Name                Variance Std.Dev. Corr
## Word     (Intercept)         3.891e-03 0.062376
```

```

##          cConcValCombo 3.584e-05 0.005986 1.00
## ID.true (Intercept) 2.317e-02 0.152229
##          cConcValCombo 1.126e-04 0.010611 -0.34
## Residual          7.823e-02 0.279699
## Number of obs: 4463, groups: Word, 40; ID.true, 38
##
## Fixed effects:
##          Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  6.591965   0.026944 47.761937 244.655  <2e-16 ***
## cConcValCombo -0.013787   0.009725 38.831847  -1.418   0.164
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##          (Intr)
## cConcValCmb 0.014
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
m = lmer(LogReactionTime ~ cValence + (1+cValence|ID.true) + (1|Word), data=val)
summary(m)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: LogReactionTime ~ cValence + (1 + cValence | ID.true) + (1 |
## Word)
## Data: val
##
## REML criterion at convergence: 1473.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -5.8193 -0.6083 -0.1559  0.4083  5.3118
##
## Random effects:
## Groups   Name            Variance Std.Dev. Corr
## Word     (Intercept) 0.003291 0.05737
## ID.true  (Intercept) 0.023454 0.15315
##          cValence    0.005980 0.07733  -0.46
## Residual                0.076980 0.27745
## Number of obs: 4463, groups: Word, 40; ID.true, 38
##
## Fixed effects:
##          Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  6.59261   0.02678 46.00933 246.202  <2e-16 ***
## cValence     -0.05837   0.02359 53.63558  -2.475   0.0165 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##          (Intr)
## cValence -0.228

```